$$\pi \int_{\pi}^{0} a^3 (1+\cos\theta)^2(1-\cos^2\theta)(1+2\cos\theta)\sin\theta \, d\theta$$

$$-a^3\pi \int_{\pi}^{0} (1+t)^2(1-t^2)(1+2t)(-dt)$$

put $\cos\theta = t$

$-\sin\theta\, d\theta = dt$

$\sin\theta\, d\theta = dt.$

$$\Rightarrow a^3\pi \int_{-1}^{1} (1+t^2+2t)(1+2t-t^2-2t^3)\,dt$$

$$\Rightarrow a^3\pi \int_{-1}^{1} 1+4t-\cancel{t^2}-\cancel{2t^3}+\cancel{t^2}+2\cancel{t^3}-t^4-2t^5 + 4t^2-2t^3-4t^4)\,dt$$

$$\Rightarrow a^3\pi \int_{-1}^{1} (1+4t+4t^2-2t^3-5t^4-2t^5)\,dt$$

$$\Rightarrow a^3\pi \left[t+\frac{4t^2}{2}+\frac{4t^3}{3}-\frac{2t^4}{4}-\frac{5t^5}{5}-\frac{2t^6}{6}\right]_{-1}^{1}$$

$$\Rightarrow a^3\pi \left[t+2t^2+\frac{4}{3}t^3-\frac{t^4}{2}-t^5-\frac{t^6}{3}\right]_{-1}^{1}$$

$$\Rightarrow a^3\pi\left(\left[\cancel{1}+2+\frac{4}{3}-\frac{1}{2}-\cancel{1}-\frac{1}{3}\right]-\left[-\cancel{1}+2-\frac{4}{3}-\frac{1}{2}+\cancel{1}-\frac{1}{3}\right]\right)$$

$$a^3\pi\left[\left(\frac{5}{2}+1\right)-\left(\frac{5}{2}-\frac{5}{3}\right)\right]$$

$$a^3\pi\left[\cancel{\frac{5}{2}}+1-\cancel{\frac{5}{2}}+\frac{5}{3}\right] = \frac{8\pi a^3}{3}$$

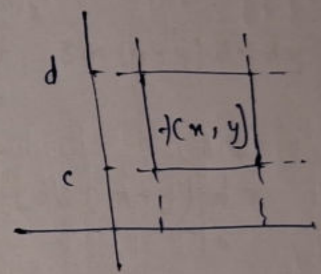
Double Integral

- $\iint f(x,y)\,dx\,dy$

$\iiint f(x,y,z)\,dx\,dy\,dz$

$f(x,y) \in$ Riemann Integrable function

$$\int_c^d \int_a^b f(x,y)\,dx\,dy$$



$$\int_a^b \int_{y=\phi_1(x)}^{y=\phi_2(x)} f(x,y)\,dy\,dx$$

$$\int_c^d \int_{x=\phi_1(y)}^{\phi_2(y)} f(x,y)\,dx\,dy$$

Q1 $\iint_R x^2 y \, dx\, dy$

$R : \{ 0 \le x \le 1 \; ; \; 0 \le y \le 2 \}$

Sol

$$\int_0^1 \int_0^2 x^2 y \, dy\, dx$$

$$\int_0^1 x^2 \times \left[\frac{y^2}{2} \right]_0^2 dx$$

$$2 \int_0^1 x^2 \, dx.$$

$$2 \left[\frac{x^3}{3} \right]_0^1$$

$$\frac{2}{3} \text{ ans}$$

Q2 $\iint_R \frac{dx\, dy}{x^2+y^2}$

$R : \{ 3 \le x \le 4 \; ; \; 1 \le y \le 2 \}$

$$\int_3^4 \int_1^2 \frac{dy}{x^2+y^2} \, dx.$$

$$\int_3^4 dx \times \left[\frac{1}{x} \tan^{-1}\left(\frac{y}{x} \right) \right]_1^2$$

$$\int_3^4 dx \times \frac{1}{x} \left[\tan^{-1}\left(\frac{2}{x} \right) - \tan^{-1}\left(\frac{1}{x} \right) \right]$$

$$\int_3^4 \frac{dx}{x} \left[\tan^{-1}\left(\frac{2}{x} \right) - \tan^{-1}\left(\frac{1}{x} \right) \right]$$

$$\int_3^4 \frac{\tan^{-1}\left(\frac{2}{x} \right)}{x} dx - \int_3^4 \frac{\tan^{-1}\left(\frac{1}{x} \right)}{x} dx$$

Q $\iint \sin(x+y) \, dx\, dy$

$R = \{ 0 \le x \le \frac{\pi}{2} \; ; \; 0 \le y \le \frac{\pi}{2} \}$

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin(x+y) \, dx\, dy.$$

$$-\int_0^{\pi/2} \left[\cos(x+y) \right]_0^{\pi/2} dx$$

$$-\left[\int_0^{\pi/2} \cos\left(\frac{\pi}{2} + x \right) dx - \int_0^{\pi/2} \cos(x) \, dx \right]$$

$$\int_0^{\pi/2} \sin(x) \, dx + \int_0^{\pi/2} \cos x \, dx.$$

$$\left[-\cos(x) \right]_0^{\pi/2} + \left[\sin x \right]_0^{\pi/2}$$

$$1 + 1$$

$$= 2 \text{ ans}$$

$$\int_0^1 \int_{y=0}^{y=\sqrt{1+x^2}} \frac{dx\,dy}{(1+x^2)+y^2}$$

Sol

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{(\sqrt{1+x^2})^2+y^2}\,dy\cdot dx$$

$$\int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1}\left(\frac{y}{\sqrt{1+x^2}}\right)\right]_0^{\sqrt{1+x^2}} dx$$

$$\int_0^1 \frac{1}{\sqrt{1+x^2}} \left[\tan^{-1}(1) - \tan^{-1}(0)\right] dx$$

$$\int_0^1 \frac{1}{\sqrt{1+x^2}} \times \frac{\pi}{4}\, dx$$

$$\frac{\pi}{4} \int_0^1 \frac{1}{\sqrt{1+x^2}}\, dx$$

$$\frac{\pi}{4} \int_0^1 \frac{1}{\sqrt{1+x^2}}\, dx$$

let $x = \tan t$

$dx = \sec^2 t\, dt$

$$\frac{\pi}{4} \int_0^{\pi/4} \frac{\sec^2 t}{\cancel{\sec t}}\, dt$$

$$\frac{\pi}{4} \int_0^{\pi/4} \sec t\, dt$$

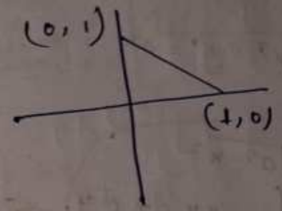
$$\frac{\pi}{4} \left[\log|\sec x + \tan x|\right]_0^{\pi/4}$$

$$\frac{\pi}{4} \left[\log[\sqrt{2}+1] - \log(1+0)\right]$$

$$\frac{\pi}{4} \log(1+\sqrt{2}) \quad a$$

Evaluate in 1^{st} quadrant where $x+y \le 1$

$$\iint_R xy\,dx\,dy$$



$$\int_0^1 \int_0^{1-x} xy\,dx\,dy$$

$$\int_0^1 \int_0^{1-x} xy\cdot dy\cdot dx$$

$$\int_0^1 x \times \left[\frac{y^2}{2}\right]_0^{1-x} dx$$

$$\frac{1}{2} \int_0^1 x(1-x)^2\, dx$$

$$\frac{1}{2} \int_0^1 x(1+x^2-2x)\, dx$$

$$\frac{1}{2} \int_0^1 x + x^3 - 2x^2\, dx$$

$$\frac{1}{2} \left[\left(\frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{3}\right]_0^1 \cancel{dx}$$

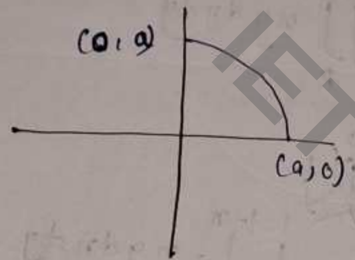
$$\frac{1}{2}\left[\frac{1}{2}+\frac{1}{4}-\frac{2}{3}\right]$$

$$\frac{1}{2}\left[\frac{6+3-8}{12}\right]$$

$$\frac{1}{2}\left[\frac{9-8}{12}\right]$$

$$=\frac{1}{24}$$

② $x^2+y^2=a^2$



$$\int_0^a\int_0^{\sqrt{a^2-x^2}} xy\,dx\,dy$$ given

$$\int_0^a\int_0^{\sqrt{a^2-x^2}} x\times\frac{y^2}{2}\,dx$$

$$\frac{1}{2}\int_0^a x(a^2-x^2)\,dx$$

$$\frac{1}{2}\left[a^2\frac{x^2}{2}-\frac{x^4}{4}\right]_0^a$$

$$\frac{1}{2}\left[a^2\times\frac{a^2}{2}-\frac{a^4}{4}\right]$$

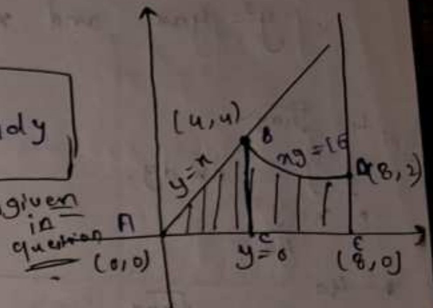
$$\frac{1}{2}\left[\frac{a^4}{4}\right]$$

$$=\frac{a^4}{8}$$ ae

Q Evaluate intrigral in 1st quad bounded by ~~$\iint x$~~ hyperbola $xy=16$ and lines $y=x$; $y=0$ and $x=8$

Sol ABC + BCDE

$$\boxed{\int_0^4\int_0^x x^2\,dx\,dy+\int_4^8\int_0^{16/x} x^2\,dx\,dy}$$



given in question

$$\int_0^4 x^2[y]_0^x\,dx$$

$$\int_0^4 x^3\,dx=\left[\frac{x^4}{4}\right]=64$$

$$\int_4^8 x^2[y]_0^{16/x}\,dx$$

$$\int_4^8 16x\,dx.$$

$$[8x^2]_4^8$$

$$8[8\times8-16]$$

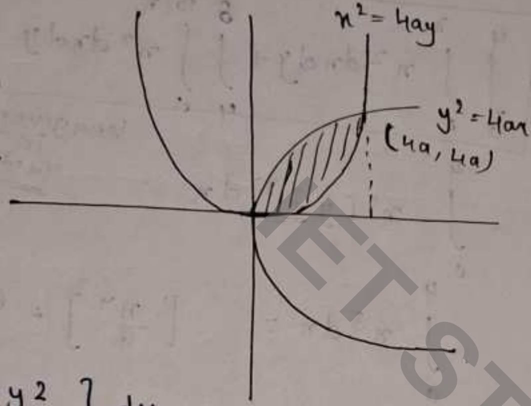
$$8\times(64-16)$$

$$=384$$

$$384+64=448$$

Q find the area of region by double intigration bounded by.

$$y^2 = 4ax \quad \text{and} \quad x^2 = 4ay$$



Sol

$$\int_0^{4a} \int_{x=\frac{y^2}{4a}}^{\sqrt{4ay}} dx\,dy.$$

$$\int_0^{4a} [x]_{\frac{y^2}{4a}}^{\sqrt{4ay}} dy$$

$$\int_0^{4a} \left[\sqrt{4ay} - \frac{y^2}{4a}\right] dy$$

$$2\sqrt{a} \times \frac{2}{3}\left[(y)^{3/2}\right]_0^{4a} - \frac{1}{4a} \times \left[\frac{y^3}{3}\right]_0^{4a}$$

$$\frac{4\sqrt{a}}{3}\left[(4a)^{3/2}\right] - \frac{1}{12a}\left[(4a)^3 - (0)^3\right]$$

$$\frac{4\sqrt{a}}{3} \times (4)^{3/2} \times a\sqrt{a} - \frac{1}{12\cancel{a}} \times 64a^{\cancel{3}2}$$

$$\frac{4a^2}{3} \times 8 - \frac{1}{\cancel{12}_{3}} \times \cancel{64}^{16} a^2$$

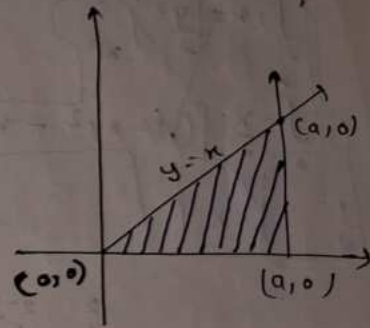
$$\frac{32a^2}{3} - \frac{16a^2}{3}$$

$$\frac{16a^2}{3}$$

Change of order of intrigation.

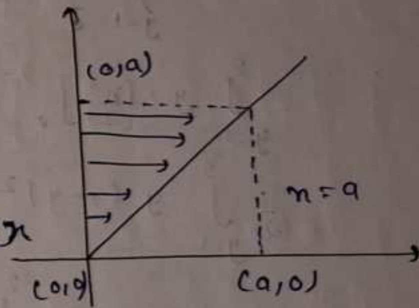
$$\int_{x=0}^{x=a} \int_{y=0}^{y=x} f(x,y)\,dx\,dy$$

$$\int_{y=0}^{y=a} \int_{x=y}^{x=a} f(x,y)\,dx\,dy$$



* $$\int_0^a \int_y^a \frac{x}{x^2+y^2}\,dx\,dy$$

$$\int_0^a \int_0^x \frac{x}{x^2+y^2}\,dx\,dy$$



$$\int_0^a \left[\cancel{x} \times \frac{1}{\cancel{x}} \tan^{-1}\left(\frac{y}{x}\right)\right]_0^x dx$$

$$\int_0^a \frac{\pi}{4}\,dx$$

$$\frac{\pi}{4}[x]_0^a$$

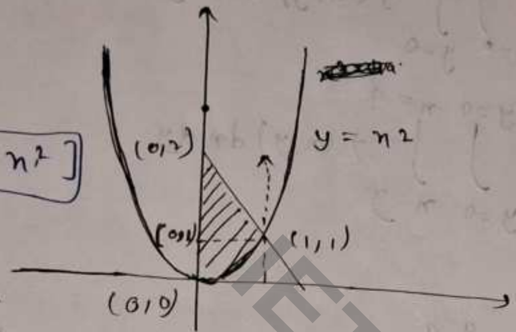
$$\frac{\pi}{4}a$$

Q $\int_0^1 \int_{x^2}^{2-x} xy \, dx\, dy$

$\int_0^1 x \times \left[\frac{y^2}{2}\right]_{x^2}^{2-x}$

$\int_0^1 \frac{x}{2} \left[(2-x)^2 - x^2\right]$

$\frac{1}{2} \int_0^1 x \, [$



$\int_{y=1}^{2} \int_0^{2-y} xy \, dx\, dy$

$\int_{y=1}^{2} y \times \left[\frac{x^2}{2}\right]_0^{2-y}$

$\frac{1}{2} \int_{y=1}^{2} y\,(2-y)^2 = \frac{1}{2} \int_{y=1}^{2} y\,(4+y^2-4y)$

$= \frac{1}{2} \int_{y=1}^{2} 4y + y^3 - 4y^2$

$= \frac{1}{2} \left[\frac{4y^2}{2} + \frac{y^4}{4} - \frac{4y^3}{3} \right]_1^2$

$= \frac{1}{2} \left[2\times 4 + \frac{16}{4} - \frac{4}{3}\times 8 - 2 - \frac{1}{4} + \frac{4}{3} \right]$

$= \frac{1}{2} \left[12 - \frac{32}{3} - 2 - \frac{1}{4} + \frac{4}{3} \right]$

$= \frac{1}{2} \left[10 - \frac{1}{4} - \frac{28}{3} \right]$

$= \frac{1}{2} \left[\frac{120 - 3 - 28\times 4}{12} \right]$

$= \frac{117 - 112}{24}$

$= \frac{5}{24}$

$\int_0^1 \int_0^{\sqrt{y}} xy \, dx\, dy$.

$\int_0^1 y \times \left[\frac{x^2}{2}\right]_0^{\sqrt{y}}$

$\frac{1}{2} \int_0^1 y \times y = \frac{1}{2} \times \int_0^1 y^2$

$= \frac{1}{2} \times \frac{y^3}{3} = \frac{1}{6}$.

$= \frac{5}{24} + \frac{1}{6} = \frac{5+4}{24} = \frac{9}{24}$

$= 3/8$

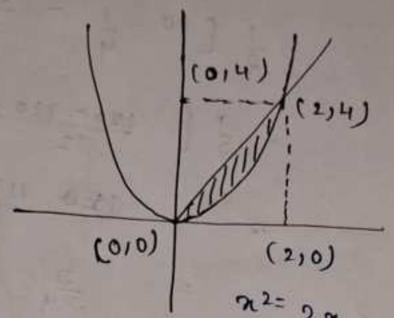
Q find Volume of the function.

$f(x,y) = 3x^2 - 3xy^2$

b/w $y = x^2$ and $y = 2x$

$$\int_0^2 \int_{x^2}^{2x} f(x,y)\,dx\,dy$$

$$\int_0^4 \int_{y/2}^{\sqrt{y}} f(x,y)\,dx\,dy.$$



$x^2 = 2x$

$x = 2$

$x = 0$

$(0,0)$, $(2,4)$

$$\int_0^2 \int_{x^2}^{2x} 3x^2 - 3xy^2$$

$$\int_0^2 \left[3x^2 y - \frac{3x}{3} y^3\right]_{x^2}^{2x}$$

$$\int_0^2 3x^2[2x - x^2] - \int_0^2 x(8x^3 - x^6)$$

$$\int_0^2 6x^3 - 3x^4 - 8x^4 - x^7$$

$$\int_0^2 6x^2 - 11x^4 - x^7$$

$$\left[\frac{6x^3}{3} - \frac{11x^5}{5} - \frac{x^8}{8}\right]_0^2$$

$$2\times 8 - \frac{11\times(2)^5}{5} - \frac{(2)^8}{8}$$

$$16 - \frac{352}{5} - 32$$

$$-16 - \frac{352}{5} = -\frac{432}{5}$$

$$\iiint_R (x+y+z)\,dx\,dy\,dz$$

$$R = \{0 \le x \le 1 ;\ 1 \le y \le 2 ;\ 2 \le z \le 3\}$$

$$\iint \left[(x+y)z + \frac{z^2}{2}\right]_2^3$$

$$\iint (x+y) + \left(\frac{9}{2} - 2\right)$$

$$\iint (x+y) + \frac{5}{2}$$

$$\int \left[xy + \frac{y^2}{2} + \frac{5}{2}y\right]_1^2$$

$$\int x + \left[\frac{4}{2} - \frac{1}{2}\right] + \left(\frac{10}{2} - \frac{1}{2}\right]$$

$$\int x + \frac{3}{2} + \frac{9}{2}$$

$$\int x + \frac{12}{2}$$

$$\int x + 6$$

$$\left[\frac{x^2}{2} + 6x\right]_0^1$$

$$\frac{1}{2} + 6 = 13/2$$

$$\int_0^{\log 2}\int_0^{x}\int_0^{x+y} e^{x+y+z}\,dx\,dy\,dz$$

Sol

$$\int_0^{\log 2}\int_0^{x}\int_0^{x+y} e^{x+y}\cdot e^{z}$$

$$\int_0^{\log 2}\int_0^{x}\left[e^{x+y}\cdot e^{z}\right]_0^{x+y}$$

$$\int_0^{\log 2}\int_0^{x} e^{x+y}\left(e^{x+y}-1\right)$$

$$\int_0^{\log 2}\int_0^{x}\left[e^{x+y}\right]^2-e^{x+y}$$

$$\int_0^{\log 2}\int_0^{x} e^{2x}\cdot e^{2y}-e^{x}\cdot e^{y}$$

$$\int_0^{\log 2} e^{2x}\cdot\left[\frac{e^{2y}}{2}\right]_0^{x}-\int_0^{\log 2} e^{x}\cdot\left[e^{y}\right]_0^{x}$$

$$\frac{1}{2}\int_0^{\log 2} e^{2x}\cdot e^{2x}-\frac{1}{2}\int_0^{\log 2} e^{2x}-\int_0^{\log 2} e^{2x}+\int_0^{\log 2} e^{x}$$

$$\frac{1}{2}\left[\frac{e^{4x}}{4}\right]_0^{\log 2}-\frac{1}{2}\left[\frac{e^{2x}}{2}\right]_0^{\log 2}-\left[\frac{e^{2x}}{2}\right]_0^{\log 2}$$

$$+\left[e^{x}\right]_0^{\log 2}$$

$$\frac{1}{4\times 2}\left[e^{4\log 2}\right]-\frac{1}{8}-\frac{1}{2\times 2}\left[e^{2\log 2}\right]+\frac{1}{4}$$

$$-\frac{1}{2}\left[e^{2\log 2}\right]+\frac{1}{2}+e^{\log 2}-1$$

$$\frac{1}{8}\times 2^4-\frac{1}{8}-\frac{1}{4}\times 2^2+\frac{1}{4}-\frac{1}{2}\times 2^2+\frac{1}{2}+2-1$$

$$2\,\cancel{\frac{16}{8}}-\frac{1}{8}-\frac{1}{\cancel{4}}\times\cancel{4}+\frac{1}{4}-\frac{1}{\cancel{2}}\times\cancel{4}^{\,2}+\frac{1}{2}+1$$

$$\cancel{2}-\frac{1}{8}-\cancel{1}+\frac{1}{4}-\cancel{2}+\frac{1}{2}+\cancel{1}$$

$$\frac{1}{2}+\frac{1}{4}-\frac{1}{8}$$

$$\frac{4+2-1}{8}=5/8 \text{ Ans}$$

Q Compute

$\iiint \dfrac{dx\,dy\,dz}{(x+y+z+1)^3}$ of the region is bounded by coordinate plane and the plane $x+y+z=1$

Sol

$0\le z\le 1-x-y$

$0\le y\le 1-x$

$0\le x\le 1$

$$\int_0^1\int_0^{1-x}\int_0^{1-x-y}\frac{dx\,dy\,dz}{((x+y+1)+z)^3}$$

$$\int_0^1\int_0^{1-x}\left(\frac{-1}{2}\frac{dx\,dy}{[(x+y+1)+z]^2}\right)_0^{1-x-y}$$

$$-\frac{1}{2}\int_0^1\int_0^{1-x} dx\,dy\left[\frac{1}{[(\cancel{x}+\cancel{y}+1)+(1-\cancel{x}-\cancel{y})]^2}-\frac{1}{(x+y+1)^2}\right]$$

$$-\frac{1}{2}\int_0^1\int_0^{1-x} dx\,dy\left(\frac{1}{(2)^2}-\frac{1}{(x+y+1)^2}\right)$$

$$-\frac{1}{2}\int_0^1\int_0^{1-x}\left[\frac{1}{4}-\frac{1}{(x+y+1)^2}\right]dx\,dy$$

$$-\frac{1}{2}\int_0^1\left[\frac{1}{4}y+\frac{1}{[(x+1)+y]}\right]_0^{1-x}dx$$

$$-\frac{1}{2}\int_0^1\left[\left(\frac{1}{4}(1-x)+\frac{1}{(\cancel{x}+1+1-\cancel{x})}\right)-\frac{1}{x+1}\right]dx$$

$$-\frac{1}{2}\int_0^1\left[\left(\frac{1}{4}-\frac{x}{4}+\frac{1}{2}\right)-\frac{1}{x+1}\right]$$

$$-\frac{1}{2}\int_0^1\frac{3}{4}-\frac{x}{4}-\frac{1}{x+1}$$

$$-\frac{1}{2}\left[\frac{3}{4}x-\frac{x^2}{8}-\log(x+1)\right]_0^1$$

$$-\frac{1}{2}\left[\frac{3}{4}-\frac{1}{8}-\log(2)\right]$$

$$-\frac{1}{2}\left[\frac{6-1}{8}-\log 2\right]$$

$$=-\frac{1}{2}\left[\frac{5}{8}-\log 2\right]$$

$$=\frac{1}{2}\log 2-\frac{5}{16}$$

Q. Find the volume of the solid bounded by parabolic $y^2+z^2=4x$ and the plane from $x=0$ to $x=5$

Sol

$$0\le z\le\sqrt{4x-y^2}$$

$$0\le y\le 2\sqrt{x}$$

$$0\le x\le 5$$

$$\int_0^5\int_0^{2\sqrt{x}}\int_0^{\sqrt{4x-y^2}} dx\,dy\,dz$$

$$\int_0^5 \int_0^{2\sqrt{x}} \int_0^{\sqrt{4x-y^2}} dx\,dy\,dz.$$

$$\int_0^5 \int_0^{2\sqrt{x}} [z]_0^{\sqrt{4x-y^2}}$$

$$\int_0^5 \int_0^{2\sqrt{x}} \sqrt{4x-y^2}$$

$$\int_0^5 \int_0^{2\sqrt{x}} \sqrt{(2\sqrt{x})^2-y^2}$$

$$\int_0^5 dx \times \frac{1}{2}\left[y\sqrt{(2\sqrt{x})^2-y^2} + 4x \sin^{-1}\left(\frac{y}{2\sqrt{x}}\right)\right]_0^{2\sqrt{x}}$$

$$\int_0^5 dx \times \frac{1}{2}\left[2\sqrt{x} \times 0 + \cancel{4}^{2} x \times \pi/\cancel{2} \right]$$

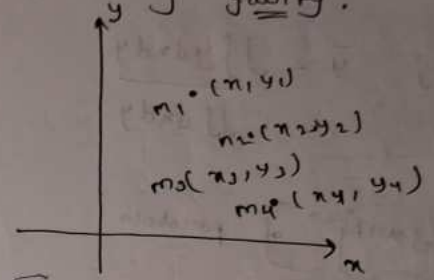
$$\frac{1}{2}\int_0^5 2x\pi$$

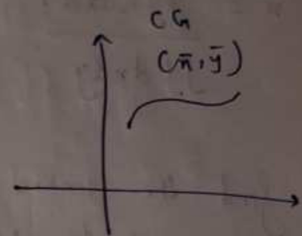
$$\frac{\cancel{2}\pi}{\cancel{2}} \int_0^5 x$$

$$\pi \times \left[\frac{x^2}{2}\right]_0^5$$

$$= \frac{25\pi}{2} \quad a$$

Centre of gravity.





$$\bar{x} = \frac{\int x\,dm}{\int dm}$$

$$\bar{x} = \frac{\int x \rho\,dv}{\int \rho\,dv}$$

$$\bar{x} = \frac{\int x \rho\,ds}{\int \rho\,ds} \qquad \bar{y} = \frac{\int y \rho\,ds}{\int \rho\,ds}$$

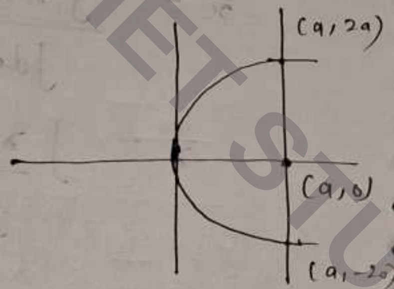
$$\bar{x} = \frac{\int x \rho\,dA}{\int \rho\,dA} \qquad \bar{y} = \frac{\int y \rho\,dA}{\int \rho\,dA}$$

$$\bar{x} = \frac{\int x \rho\,dv}{\int \rho\,dv} \qquad \bar{y} = \frac{\int y \rho\,dv}{\int \rho\,dv}$$

$$\bar{x} = \frac{\iint x\,dx\,dy}{\iint dx\,dy} \quad ; \quad \bar{y} = \frac{\iint y\,dx\,dy}{\iint dx\,dy}$$

Q. Find the centre of gravity of parabola bounded by latus rectum

Parabola $y^2 = 4ax$



Sol

$$\bar{x} = \frac{\int_0^a x \rho\, dA}{\int \rho\, dA}$$

$$= \frac{\int x \cancel{\rho}(y\,dx)}{\int \cancel{\rho} \times y\,dx} \Rightarrow \frac{\int_0^a xy\,dx}{\int_0^a y\,dx}$$

$$\Rightarrow \frac{\int_0^a x \times 2\sqrt{a}\cdot\sqrt{x}\,dx}{\int_0^a 2\sqrt{a}\cdot\sqrt{x}\,dx} \Rightarrow \frac{\int_0^a x\sqrt{x}\,dx}{\int_0^a \sqrt{x}\,dx}$$

$$\frac{\left[\frac{2}{5}(x)^{5/2}\right]_0^a}{\left[\frac{2}{3}(x)^{3/2}\right]_0^a} = \frac{\frac{\cancel{2}}{5} \times a^{5/2}}{\frac{\cancel{2}}{3}(a)^{3/2}}$$

$$\bar{x} = \frac{3}{5}a$$

$$\bar{y} = \cancel{\iint y}\ \frac{\int y \rho\, ds}{\int \rho\, ds}$$

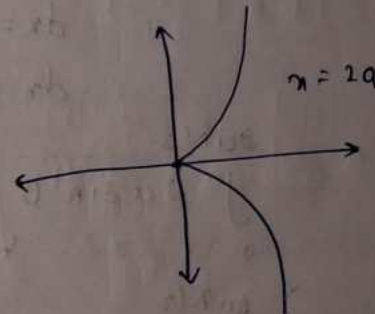
$$= \frac{\int y \cancel{\rho} \times x\,dy}{\int \cancel{\rho} \times x\,dy}$$

$$= \frac{\int_0^{2a} xy\,dy}{\int x\,dy}$$

Centre of gravity $\left(\frac{3}{5}a, 0\right)$

→ Find the centre of gravity of the area included b/w the curve $y^2(2a-x) = x^3$ and its asymptotes.

$$\bar{x} = \frac{\int x\,dm}{\int dm}$$



$$\bar{y} = \frac{\int y\,dm}{\int dm}$$

$\bar{y} = 0$

$dm = \rho\, dA$

$= \rho(2y\,dx$

$$\bar{x} = \frac{\int_0^{2a} x\rho\,(2y\,dx)}{\int_0^{2a} \rho\,(2y\,dx)}$$

$$\Rightarrow \frac{\int_0^{2a} xy\,dx}{\int_0^{2a} y\,dx} = \frac{\int_0^{2a} x\cdot\sqrt{\frac{x^3}{2a-x}}\,dx}{\int_0^{2a}\sqrt{\frac{x^3}{2a-x}}\,dx}.$$

$$\int_0^{2a} x\sqrt{\frac{x^3}{2a-x}}\,dx$$

let $x = 2a\sin^2\theta$

$dx = 2a \times 2\sin\theta\cos\theta\, d\theta$

$dx = 4a\sin\theta\cos\theta$

$$\int_0^{\pi/2} 2a\sin^2\theta\sqrt{\frac{8a^3\sin^6\theta}{2a-2a\sin^2\theta}} \times 4a\sin\theta\cos\theta\, d\theta$$

$$8a^2\int_0^{\pi/2}\sqrt{\frac{4\,8a^{3\,2}\sin^6\theta}{2a\cos^2\theta}} \times \sin^3\theta\cos\theta\, d\phi$$

$$8a^2 \times 2a\int_0^{\pi/2}\sin^3\theta\cdot\sin^3\theta\, d\theta$$

$$16a^3\int_0^{\pi/2}\sin^6\theta\, d\theta$$

$$\frac{16a^3}{2}\,\frac{\Gamma\frac{6+1}{2}\,\Gamma\frac{1}{2}}{\Gamma\frac{6+0+2}{2}}$$

$$8a^3\,\frac{\Gamma\frac{7}{2}\,\Gamma\frac{1}{2}}{\Gamma 4}$$

$$\frac{8a^3}{6} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}\times\sqrt{\pi}$$

$$\frac{15a^3\pi}{6} = \frac{5}{2}a^3\pi$$

$$\rightarrow \int_0^{2a}\sqrt{\frac{x^3}{2a-x}}\,dx$$

let $x = 2a\sin^2\theta$

$dx = 2a \times 2\sin\theta\cos\theta\, d\theta$

$= 4a\sin\theta\cos\theta\, d\phi$

$$\int_0^{\pi/2}\sqrt{\frac{8a^3\sin^6\theta}{2a-2a\sin^2\theta}} \times 4a\sin\theta\cos\theta\, d\phi$$

$$= \int_0^{\pi/2}\sqrt{\frac{4\,8a^{3\,2}\sin^6\theta}{2a\cos^2\theta}} \times 4a\sin\theta\cos\phi$$

$$8a^2 \int_0^{\pi/2} \sin^3\theta \sin\theta \, d\theta$$

$$8a^2 \int_0^{\pi/2} \sin^4\theta \, d\theta$$

$$\frac{8a^2}{2} \frac{\Gamma 5/2 \, \Gamma 1/2}{\Gamma 6/2}$$

$$2 \frac{\cancel{8}a^2}{\cancel{2}\times\cancel{2}} \times \frac{3}{2}\times\frac{1}{2}\sqrt{\pi}\sqrt{\pi}$$

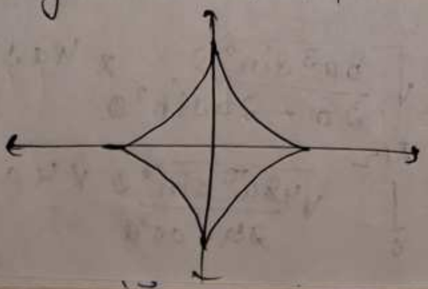
$$\frac{\cancel{2}a^2\times 3}{\cancel{2}\times 2}\pi$$

$$\frac{3a^2}{2}\pi$$

$$\frac{\frac{5a^3}{\cancel{2}}\cancel{\pi}}{\frac{3a^2}{\cancel{2}}\cancel{\pi}} = \frac{5a}{3}$$

COG $\left(\frac{5a}{3}, 0\right)$

Q Find COG of the area bounded in tve ~~cor~~ quadrant ~~[illegible]~~ of the astroid



$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$\bar{x} = \frac{\iint x\,dx\,dy}{\iint dx\,dy}$$

$$\bar{y} = \frac{\iint y\,dx\,dy}{\iint dx\,dy}$$

$$\bar{x} = \frac{\int_0^a \int_0^{(a^{2/3}-x^{2/3})^{3/2}} x\,dx\,dy}{\int_0^a \int_0^{(a^{2/3}-x^{2/3})^{3/2}} dx\,dy}$$

$$= \frac{\int_0^a x\,dx\left([a^{2/3}-x^{2/3}]^{3/2}\right)}{\int_0^a (a^{2/3}-x^{2/3})^{3/2}}$$

let $x = a\sin^3\theta$

$dx = a \times 3\sin^2\theta\cos\theta\, d\theta$

$$\int_0^{\pi/2} a\sin^3\theta\,(a^{2/3} - a^{2/3}\sin^2\theta)^{3/2} \times 3a\sin^2\theta \cos\theta\, d\theta$$

$$I = 3a^3 \int_0^{\pi/2} \sin^3\theta \,(\cos^2\theta)^{3/2} \sin^2\theta \cos\theta \, d\theta$$

$$= 3a^3 \int_0^{\pi/2} \sin^3\theta \cos^3\theta \sin^2\theta \cos\theta \, d\theta$$

$$= 3a^3 \int_0^{\pi/2} \sin^5\theta \cos^4\theta \, d\theta$$

$$= \frac{3a^3}{2} \frac{\Gamma\frac{5+1}{2}\,\Gamma\frac{4+1}{2}}{\Gamma\frac{5+4+2}{2}}$$

$$= \frac{3a^3}{\cancel{2}} \times \frac{\cancel{2} \times \frac{3}{\cancel{2}} \times \frac{1}{\cancel{2}}\sqrt{\cancel{\pi}}}{\frac{9}{\cancel{2}} \times \frac{7}{\cancel{2}} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2}\sqrt{\pi}}$$

$$= \frac{\overset{3}{\cancel{8}}a^3}{7 \times 9 \times 5 \times \cancel{3}} = \frac{3a^3}{45 \times 7}$$

$$= \frac{\cancel{3}a^3}{\underset{15}{\cancel{45}} \times 7} = \frac{3a^3}{15 \times 7}$$

$$I = \int_0^{\pi/2} (a^{2/3} - a^{2/3}\sin^2\theta)^{3/2} \times 3a\sin^2\theta \cos\theta \, d\theta$$

$$= 3a^2 \int_0^{\pi/2} \cos^4\theta \sin^2\theta \, d\theta$$

① $\vec{F} = (yz\hat{i} + zx\hat{j} + xy\hat{k})$

Evaluate surface integral over S where S is the surface of the sphere $x^2+y^2+z^2=1$ in the 1^{st} octant

Sol

$$\int \vec{F}\cdot\hat{n}\, ds.$$

$$\hat{n} = \frac{\text{grad } \vec{F}}{|\text{grad } \vec{F}|}$$

$$\text{grad } (\nabla) = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)$$

$$|\text{grad } (\nabla)| = \sqrt{\left(\frac{\partial}{\partial x}\right)^2 + \left(\frac{\partial}{\partial y}\right)^2 + \left(\frac{\partial}{\partial z}\right)^2}$$

$$\hat{n} = \frac{\nabla \vec{f}}{|\nabla f|}$$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{\sqrt{4x^2+4y^2+4z^2}}$$

$$\hat{n} = \frac{\cancel{2}x\hat{i} + \cancel{2}y\hat{j} + \cancel{2}z\hat{k}}{\cancel{2}\sqrt{x^2+y^2+z^2}} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\hat{n} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$ds = \frac{dx.dy}{(\hat{n}.\hat{k})}$$

$$= \frac{dx.dy}{(x\hat{i}+y\hat{j}+z\hat{k}).\hat{k}}$$

$$= \frac{dx.dy}{z}$$

$$\int_0^1\int \vec{F}.\vec{n}\,ds = (yz\hat{i}+zx\hat{j}+xy\hat{k}).(x\hat{i}+y\hat{j}+z\hat{k})\frac{dxdy}{z}$$

$$\int_0^1\int_0^{\sqrt{1-x^2}} (xy\cancel{z}+xy\cancel{z}+xy\cancel{z})\,\frac{dx.dy}{\cancel{z}}$$

$$3\int_0^1\int_0^{\sqrt{1-x^2}} xy\,dx.dy$$

$$\frac{3}{2}\int_0^1 x\times\left[y^2\right]_0^{\sqrt{1-x^2}}$$

$$\frac{3}{2}\int_0^1 x(1-x^2)$$

$$\frac{3}{2}\int_0^1 x-x^3$$

$$\frac{3}{2}\left[\frac{x^2}{2}-\frac{x^4}{4}\right]_0^1$$

$$\frac{3}{2}\left[\frac{1}{2}-\frac{1}{4}\right]$$

$$\frac{3}{2}\left[\frac{2-1}{4}\right]$$

$$= 3/8$$

Q $\iiint (2xz\hat{i}-x\hat{j}+y^2\hat{k})\,dv$

Calculate Volume integral where V is the region ~~by~~ bounded by the surfaces $x=0,\ y=6,\ \ ,\ y=6,\ z=x^2,\ z=4$

Sol
$$\int_0^2\int_0^6\int_{x^2}^4 (2xz\hat{i}-x\hat{j}+y^2\hat{k})\,dxdydz$$

$$\int_0^2\int_0^6 \left[\cancel{2}x\frac{z^2}{\cancel{2}}\hat{i}-xz\hat{j}+y^2z\hat{k}\right]_{x^2}^4$$

$$\int_0^2\int_0^6 \left[x\times16\hat{i}-4x\hat{j}+4y^2\hat{k}-x^5\hat{i}+x^3\hat{j}-x^2y^2\hat{k}\right]$$

$$\int_0^2\int_0^6 \hat{i}(16x-x^5)+\hat{j}(-4x+x^3)+\hat{k}(4y^2-x^2y^2)$$

$$\int_0^2\left[(16x-x^5)y\hat{i}+(-4x+x^3)y\hat{j}+\hat{k}\times\frac{y^3}{3}(4-x^2)\right]_0^6$$

$$\int_0^2 6(16x - x^5)\hat{i} + 6(-4x + x^3)\hat{j} + \frac{\cancel{6}\times 6\times 6}{\cancel{3}}(4-x^2)\hat{k}$$

$$6\left[\frac{16x^2}{2} - \frac{x^6}{6}\right]_0^2 \hat{i} + 6\left(-\frac{4x^2}{2} + \frac{x^4}{4}\right]_0^2 \hat{j} + 72\left(4x - \frac{x^3}{3}\right]_0^2$$

$$6\hat{i}\left[\frac{16}{\cancel{2}}\times\cancel{2}\times 2 - \frac{64}{6}\right] + 6\hat{j}\left[-2\times2\times2 + 4\right] + 72\left[8 - \frac{8}{3}\right]$$

$$6\hat{i}\left(32 - \frac{32}{3}\right] + 6\hat{j}(-4) + \frac{72}{3}(16)\hat{k}$$

$$2\cancel{6}\times 32\,\hat{i}\left(\frac{2}{\cancel{3}}\right) + 0 - 24\hat{j} + 24\times 16\,\hat{k}$$

$$\vec{v} = 128\hat{i} - 24\hat{j} + 384\hat{k}$$

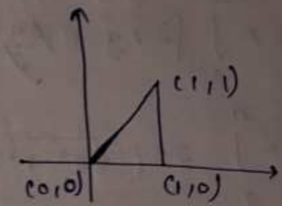
* <u>Green's theorem</u>

If $\phi(x,y)$, $\psi(x,y)$, $\frac{\partial\phi}{\partial y}$, $\frac{\partial\psi}{\partial x}$ be continuous function over a region are bounded by simple closed curve C in xy plane then

$$\oint_C(\phi\,dx + \psi\,dy) = \iint_R\left(\frac{\partial\psi}{\partial x} - \frac{\partial\phi}{\partial y}\right)dx\,dy$$

Q Using green's theorem where C is the boundary described Counter clock wise of triangle with vertices $(0,0)$ $(1,0)$ and $(1,1)$

Sol $\oint_C x^2y\,dx + x^2\,dy$



$$\int_0^1\int_0^x (2x - x^2)\,dx\,dy.$$

$$\int_0^1 \left[\frac{\cancel{2}x^2}{\cancel{2}} - \frac{x^3}{3}\right]_0^x$$

$$\int_0^1 \left[x^2 - \frac{x^3}{3}\right]dx$$

$$\left[\frac{x^3}{3} - \frac{x^4}{12}\right]_0^1$$

$$\left[\frac{1}{3} - \frac{1}{12}\right]$$

$$= \frac{\quad}{12}$$

$$\int_0^1 \left[(2x - x^2)y\right]_0^x$$

$$\int_0^1 2x^2 - x^3$$

$$\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^1$$

$$\left[\frac{2}{3} - \frac{1}{4}\right]$$

$$= \frac{8-3}{12} = \frac{5}{12}\ \text{units}$$

Q Apply green's theorem to evaluate

$$\int_C (2x^2 - y^2)\,dx + (x^2 + y^2)\,dy$$

Curve C is the boundary of the area enclosed by x-axis and the upper half of the circle $x^2 + y^2 = a^2$

Sol

$$2\times\int_0^a\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} (2x+2y)\,dx\,dy$$

$$2\times\int_0^a \left[2xy+\cancel{2}\frac{y^2}{\cancel{2}}\right]_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}$$

$$2\times\int_0^a 2x\left(\sqrt{a^2-x^2}+\sqrt{a^2-x^2}\right)+\left(\cancel{a^2}-\cancel{x^2}-\cancel{a^2}+\cancel{x^2}\right)$$

$$2\times\int_0^a 2x\times 2\sqrt{a^2-x^2}$$

$$2\times 4\int_0^a x\sqrt{a^2-x^2}$$

let $x=a\sin\theta$

$dx=a\cos\theta\,d\theta$

$$2\times 4\int_0^{\pi/2} a\sin\theta\sqrt{a^2-a^2\sin^2\theta}\times a\cos\theta\,d\theta$$

$$2\times 4a^2\int_0^{\pi/2}\sin\theta\cos^2\theta\,d\theta$$

$$2\times 2a^2\,\frac{\Gamma\frac{1+1}{2}\,\Gamma\frac{2+1}{2}}{\Gamma\frac{2+1+2}{2}} \Rightarrow \frac{2a^2\,\Gamma 1\,\Gamma\frac{3}{2}}{\Gamma 5/2}$$

$$\frac{2\times 2a^2\times\frac{1}{\cancel{2}}\cancel{\sqrt{\pi}}}{\frac{3}{2}\times\frac{1}{\cancel{2}}\cancel{\sqrt{\pi}}}$$

$$=\frac{2\times 4a^3}{3}=\frac{8a^3}{3}$$

Gauss Theorem of Divergence

The surface integral of the normal component of a vector function F taken around the closed surface S is equal to the integral of the divergence of F taken over the volume V enclosed by the surface S

$$\boxed{\iint_S \vec{F}\cdot\hat{n}\,ds=\iiint_V \text{div}\,\vec{F}\,dv}$$

① Evaluate surface integral $\iint_S \vec{F}\cdot\hat{n}\,ds$ where $\vec{F}=4xz\hat{i}-y^2\hat{j}+yz\hat{k}$ bounded by $x=0,\ x=1,\ y=0,\ y=1$ and $z=0,\ z=1$

$$\iint \vec{F}\cdot\hat{n}\,ds=\iiint_V \text{div}\,\vec{F}\,dv$$

$$\text{div}\,f=\left(\cancel{4z\hat{i}-2y\hat{j}+y\hat{k}}\right)\ 4z-2y+y$$

$$=(4z-y)$$

$$\int_0^1\int_0^1\int_0^1 \cancel{(4z\hat{i}-2y\hat{j}+y\hat{k})\,dx\,dy\,dz}$$

$$\int_0^1\int_0^1\int_0^1 (4z-y)\,dx\,dy\,dz$$

$$\int_0^1\int_0^1 \frac{4z^2}{2} - yz$$

Q Apply divergence theorem to evaluate volume integral $\iiint_V \text{div}\,\vec{f}\,dv$ where vector function $\vec{f} = 4x^3\hat{i} - x^2y\hat{j} + x^2z\hat{k}$ and S is surface of the cylinder $x^2+y^2=a^2$ bounded by the planes $z=0$ and $z=b$

Sol

$$\text{div.}\,\vec{f} = 12x^2$$

$$\iint \vec{F}\cdot\hat{n}\,ds = \iiint \text{div.}\vec{f}\,dv$$

$$= \int_0^b\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} 12x^2\,dx\,dy\,dz$$

$$\int_0^b\int_{-a}^{a}\left[12x^2\cdot z\right]_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}$$

$$2\times 12\int_0^b\int_{-a}^{a} x^2\sqrt{a^2-x^2}$$

$$48\int_0^b\int_0^a x^2\sqrt{a^2-x^2}$$

~~$x = a\sin\theta$~~ ~~$x = a\tan\theta$~~

$dx = a\cos\theta\,d\theta$. ~~$dx = a\sec^2\theta\,d\theta$~~

$$48\int_0^b\int_0^{\pi/2} \cancel{a^2\sin^2\theta\sqrt{a^2+a^2\sin^2\theta}}$$

$$48\int_0^b\int_0^{\pi/2} a^2\sin^2\theta\sqrt{a^2-a^2\sin^2\theta}\;a\cos\theta\,d\theta$$

$$48a^3\int_0^b\int_0^{\pi/2} \sin^2\theta\cos^2\theta\,d\theta$$

$$\frac{48a^3}{2}\int_0^b \frac{\Gamma\frac{2+1}{2}\;\Gamma\frac{2+1}{2}}{\Gamma\frac{2+2+2}{2}}$$

$$24a^3\;\frac{\Gamma\frac{3}{2}\;\Gamma\frac{3}{2}}{\Gamma 3}\;b$$

$$\frac{24a^3b\times\frac{1}{2}\sqrt{\pi}\times\frac{1}{2}\sqrt{\pi}}{2} = \underline{\underline{3a^3b\pi}}$$

~~Stock~~ Stoke's theorem

Surface integral of the component of curl f along the normal to the surface S taken over the surface S bound by the curve C is equal to the line integral of the vector point function f taken along ~~from~~ closed curve C

Or

The line integral of the vector function f around any closed curve is equal to the surface integral of curl f taken over any surface of which the curl is a boundary ~~H~~ edge

$$\oint \vec{F}\,d\vec{r} = \iint \text{curl}\,\vec{F}\,\hat{n}\,ds$$

$$\hat{n} = \cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k}$$

$$ds = \frac{dx\,dy}{(\hat{n}\cdot\hat{k})}$$

$$\hat{n} = \frac{\nabla \vec{F}}{|\nabla \vec{F}|}$$

Q Using stoke's theorem evaluate line integral

$$\int_C (2x-y)\,dx - yz^2\,dy - y^2 z\,dz$$

where C is the circle of unit radius corresponding to the surface of sphere of unit radius

Sol

$$\text{curl } f = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2 z \end{vmatrix}$$

$$\hat{i}(-2yz + 2yz) - \hat{j}(0) + \hat{k}(0+1)$$

$$\text{curl } f = \hat{k}$$

$$\iint \hat{k}\cdot\hat{n}\,\frac{dx\,dy}{\hat{n}\cdot\hat{k}}$$

$$\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\!\!\int dx\,dy = 2\int\!\!\int_0^{\sqrt{1-x^2}} dx\,dy \Rightarrow 2\int^1 \sqrt{1-x^2}\,dx$$

$$= \pi(1)^2 \Rightarrow 2\int \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta$$

$$= \pi$$

$$= 2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$4\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 4\,\frac{\Gamma\frac{0+1}{2}\,\Gamma\frac{2+1}{2}}{2\,\Gamma\frac{2+2}{2}} = \frac{4}{2}\sqrt{\pi}\times\frac{1}{2}\sqrt{\pi} = \pi$$

~~By Stoke's theorem~~

H.W.

Q Evaluate line integral $\int \vec{f} \cdot dr$ by stoke's theorem where $\vec{f} = (x^2+y^2)\hat{i} - 2xy\hat{j}$ and rectangle $x = \pm a$, $y = 0$, $y = b$

Q Apply stoke's theorem to calculate to

$$\int_C 4y\,dx + 2z\,dy + 6y\,dz$$

where C is curve of intersection of $x^2+y^2+z^2 = 6z$ and $z = x+3$

$4\sqrt{2}$

Sol $\vec{f} = (x^2+y^2)\hat{i} - 2xy\hat{j}$

$x = \pm a$, $y = 0$, $y = b$.

$$\int \vec{f} \cdot d\vec{r} = \iint_R \text{curl} f \cdot \hat{n} \cdot ds$$

$$\text{curl} f = \nabla f \times f$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$$

$$\hat{i}(0) - \hat{j}(0) + \hat{k}(-2y-2y)$$

$$\text{curl} f = -4y\hat{k}$$

$$\int_{-a}^{a}\int_{0}^{b} -4y\cancel{\hat{k}} \cdot \cancel{\hat{n}} \cdot \frac{dx\,dy}{\cancel{\hat{n} \cdot \hat{k}}}$$

$$-4\int_{-a}^{a}\int_{0}^{b} y\,dx\,dy$$

$$-4\int_{-a}^{a} \left[\frac{y^2}{2}\right]_0^b$$

$$-\frac{4}{2}\int_{-a}^{a} b^2$$

$$-2b^2 \int_{-a}^{a} dx$$

$$-2b^2 [x]_{-a}^{a}$$

$$-2b^2 [a+a]$$

$$\underline{\underline{-4ab^2}}$$

Q $\int_C 4y\,dx + 2z\,dy + 6y\,dz$.